

Testing a Dual Water- and Space-Heating System

I tested a dual system for domestic hot water and space heating in a home where a furnace and ductwork were previously used for space heating.

by Roy H. Jensen

A high-efficiency on-demand water heater can provide both domestic hot water and space heating at a fraction of the cost of a separate tank water heater and furnace.

Hot water tanks and furnaces operate on the same principle: heating water for domestic hot water (DHW) or heating air for space heating. Both hot water tanks and furnaces are usually powered by electricity or fossil fuels (natural gas, propane, or heating oil). Electric units have the advantage of efficiencies approaching 100% in site energy, but they have disadvantages: They require large amounts of electricity during peak periods, electricity costs more than fossil fuels, and the efficiencies of the power plants that produce the electricity from fossil fuels average about 30%. Fuel costs are lower for gas- or oil-burning units, but appliance efficiency is also lower because hot exhaust gases must be vented and energy extraction from the exhaust is not 100%.

Efficiency is a key consideration when choosing new heating appliances. One should consider the unit's AFUE (annualized fuel utilization efficiency) and its EF (energy factor). The AFUE is the seasonal efficiency of a heating appliance. The EF is used to report the efficiency of a water heater and is the fraction of useful energy output per

unit energy input. Conventional space- and water-heating appliances have efficiencies ranging from 50% to 80%. High-efficiency appliances have greater than 80% AFUE or EF. (This includes both condensing heating appliances, which extract heat energy from exhaust gases, and non-condensing heating appliances.)

Appliances that function continuously, such as hot water tanks and boilers, have to maintain a ready state (in this case, a supply of hot water). This decreases the efficiency of the appliance, wastes energy, and wastes money. For example, while you are asleep, at work, on vacation, or comfortably reading this article, your hot water tank is faithfully maintaining a ready supply of hot water. The insulation on the hot water tank is not perfect (and never can be). Heat slowly and constantly bleeds from the tank into the surrounding air. Hot water tanks often keep water hotter than required to increase their capacity, resulting in greater and faster heat loss. It is obviously better to have appliances that operate only when required.

Hot water can be generated when required using an on-demand tankless water heater (ODWH). While they are not common in North America, ODWHs have been used in Europe and Asia for over 30 years. The

technology is proven and robust. A typical ODWH is 15 inches high x 24 inches wide x 12 inches deep (35 cm x 60 cm x 30 cm). It is wall mounted and operates on either natural gas or propane. Electric units are available, but these units use a great deal of electricity and often cannot supply enough hot water for more than one appliance or fixture at a time.

The Okaloosa Gas District, in Okaloosa, Florida, conducted a study in 2002 to compare on-demand and tank water heaters. The study found that a Rinnai Continuum ODWH used 45% less energy than a natural gas hot water tank (Rheem model 21V40-38) and 35% less energy than an electric hot water tank (Rheem model 81V40D). The test procedures were derived from the "Test Procedures for Water Heaters" outlined by the Federal Register, Title 10 of the Code of Federal Regulations (CFR), Section 430. These procedures provide a standard for fair comparison between energy efficiency, energy use, and the annual operating cost of water heaters.

ODWHs are superior to conventional water heaters (hot water tanks and boilers) because on-demand water heaters provide hot water only when needed; there is no wasteful hot water reservoir; since there is no reservoir, the water is not overheated to



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The author stands in front a house in which he tested a dual water heating and space heating system. Looks like winter conditions in the Canadian neighborhood.

extend reservoir supply; they provide an infinite supply of hot water (albeit at a limited flow rate); they have a life expectancy of 20 to 30 years, compared with 8 to 12 years for hot water tanks; and they cost less to buy and they cost no more to install (see Table 1).

Dual water- and space-heating systems are not new, but they have not been well received in North America, possibly due to their historical high cost, historical limited efficiency savings, historical large space requirements, and consumer and dealer resistance to change. Previous dual systems often employed multiple hot water tanks or a boiler with multiple heat exchangers to supply both DHW and space heating. These systems required substantial space, and they wasted energy by maintaining one or more reservoirs of hot water.

Existing heating technology can be improved by employing an ODWH, without a wasteful hot water reservoir, for both DHW and space heating. Instead of separate heaters in the hot water tanks and in the furnace—each with its own chimney and varying efficiencies—a dual system employs a single heater to supply hot water for domestic use and to heat air for space heating. I designed and tested a dual system for domestic hot water and space heating in a home where a furnace and ductwork were previously used for space heating.

The ODWH could equally well be used in a home that was previously heated hydronically.

Components, Features, and Operation of an ODWH

Briefly, an ODWH heats water for DHW and to supply a fan coil heat exchanger for space heating (see Figure 1). The fan coil and ODWH are installed in a closed-loop configuration, with a pump circulating the fan coil outflow into the intake of the ODWH. Domestic hot water is extracted from the loop and tempered to minimize the risk of scalding.

Table 1. Typical Appliance Prices and Efficiencies

Unit	Price/ CAN \$	AFUE or EF/%	Life Expectancy /years
Mid-efficiency furnace	800–1,500	50–80	20–30
High efficiency furnace	1,500–3,000	80+	20–30
High efficiency furnace (w/VSM*)	2,500–4,000	80+	20–30
Electric baseboard heating	1,500–2,000	95–100	20–30
Electric HWT	400– 600	85–100	10–15
Standard fuel HWT	400– 900	50–80	8–12
High efficiency HWT	1,500–2,200	80+	8–12
High efficiency boiler	4,500–6,000	80+	20–30
ODHW	1,200–2,000	80+	20–30
*Variable speed motor			

To simultaneously meet the DHW and space-heating requirements of a typical home, the ODWH should be able to heat water to 160°F (70°C) and have an energy output greater than 180,000 Btu per hour. Bosch, Noritz, Paloma, Rinnai, and Tagaki all have one or more models that meet these requirements. I used a Paloma Waiwela ODWH (model PH28CIF5; 199,900 Btu per hour; \$1,400) to test a dual system.

Like all high-efficiency condensing appliances, ODWHs require class B or stainless steel venting but can be vented horizontally. A mixture of elbows and linear segments totaling three meters costs approximately \$500. Some ODWHs employ a dual-vent system and can draw in outside air for combustion, which further increases efficiency.

Space heating can be accomplished using radiant floors, baseboards, or a fan coil connected to existing ductwork. Specially designed subfloors can accommodate hydronic lines. A heated fluid pumped through baseboards or the hydronic tubing conducts heat to

Dual System Schematic

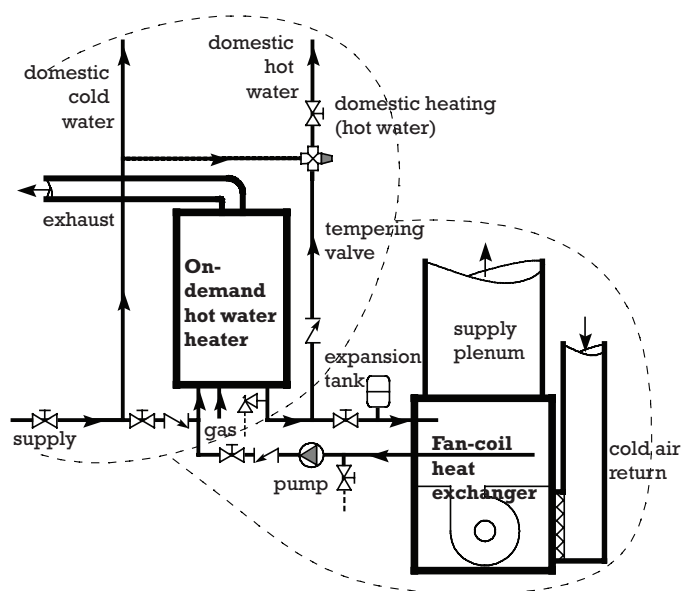


Figure 1. An ODWH heats water for DHW and to supply a fan coil heat exchanger for space heating. All water lines should be copper until beyond the tempering valve.

the room. Hydronic systems cannot be retrofitted without major costly renovations. They are, however, easy to install during the construction of new homes and they provide clean, uniform, and quieter space heating. The ODWH, through a liquid-liquid heat exchanger, is an inexpensive high-efficiency heat source for hydronic lines.

A fan coil is a liquid-to-air heat exchanger, much like an automobile radiator (conventional furnaces are air-to-air heat exchangers). The high heat capacity of water allows the fan coil to operate at lower temperatures than furnaces. Operating on natural



The installed system follows the schematic in Figure 1. The fan coil is on the other side of the cold air return. All hot water lines are insulated.

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integrated into the existing ductwork by replacing the furnace.

A fan coil space-heating system offers the following benefits: It is up to 100% efficient (no energy is wasted, since the outflow water is recycled back into the ODWH); it has no second chimney or exhaust; and it can function as a central cooling system if a cold-water supply is available.

The domestic water-heating component in the test home is typical for any domestic water heater (see Figure 1). Because the ODWH can heat water to 180°F (82°C), and higher water temperatures are required for space heating in winter, a tempering valve

(Honeywell model AM101; \$100) was used in the system to ensure that the DHW temperature stays reasonable, at 125°F (52°C). Check valves ensure the correct water flow when the space-heating component is functioning.

The space-heating component forms a closed loop and uses a circulating pump (Taco model 0011-BF4; \$350) to circulate water through the ODWH and fan coil. The pump is controlled by the fan coil and is wired to start when the thermostat requests heat. Check valves prevent the pump from circulating warm water into the domestic cold supply or from extracting water and possibly air from the DHW lines. The pump is located downstream from the heat exchanger to minimize its thermal load and extend its life. Because the space-heating system is closed loop and subject to thermal cycling, a thermal expansion tank (Diatrol model 537; \$110) is incorporated into the system

to accept thermal expansion. The check valve on the DHW line is situated a distance from the tee to provide a trap for gases that may be entrained in the water flow.

In operation, the ODWH should be set to give hot water at the lowest possible temperature, to minimize thermal stress, minimize scaling, increase efficiency, and increase the operational life of all components. During periods when the need for space heating is minimal, the ODWH should be set to 125°F (52°C), which is sufficient for DHW and space heating. With outside temperatures below 14°F (-10°C), the ODWH should be set to maintain a comfortable interior temperature with a duty cycle of less than 20%. This duty cycle was chosen to reduce the chance of insufficient domestic hot water, but all appliances are rated to function with at least a 95% duty cycle.

Seasonally adjusting the ODWH to the lowest possible temperature minimizes the rate of scale buildup in the system (see Table 2). Scale (calcium and magnesium carbonate) is less soluble in hot water than it is in cold water. If scale is present, it can be removed by flushing the ODWH and fan coil with a consumer phosphoric acid-based descaling agent, which does not damage copper. Depending on the hardness of the local water, this may have to be done as often as every five years; or it may never have to be done at all.

On start-up, opening the drain line will flush the stale water in the fan coil. During the heating season, water will not stagnate, because it is drawn off by the domestic supply.

Testing a Dual System

The test home is a 1,900-ft² (176 m²) bungalow with a fully finished basement in Edmonton, Alberta, Canada. It was built in 1959 with 4-inch (10 cm) walls. The test ODWH system has been used in the home since fall 2005. It has performed

gas, a furnace produces a maximum combustion temperature of 3,900°F (2,150°C) in half of the heat exchanger. The heat exchanger heats room air to approximately 68°F (20°C) in the other half. The high temperature differential in furnaces leads to static thermal stresses during operation and variable thermal stresses when the furnace turns on and off. Although they are purportedly designed for these extreme temperature differentials, heat exchangers often fail prematurely due to design flaws and manufacturing defects. Fan coils are commonly used with heat pumps, and in geothermal and solar heating systems. They operate at a maximum temperature of the boiling point of the heating fluid (below 212°F [100°C] for water), resulting in significantly less thermal stress. A variable-speed fan coil (Energy Saving Products model LV-120; \$1,100) was used in the test system. It was

flawlessly for a family of two adults and one youth. Set at the default temperature of 125°F (52°C), the system functioned successfully when the outside temperature was above 14°F (10°C). When the outside temperature ranged from -4°F to -40°F (-20°C to -40°C) for a three-week period, the system functioned successfully when the ODWH was set to supply water at 160°F (70°C). To date, there is no evidence of scale in the system.

The preinstallation energy usage in the test home is not known. The postinstallation energy usage (since November 2005) for space heating and hot water is 5.9 Btu per square foot per heating degree-day (HDD) (at a 65° F base; 67 megajoules [MJ] per square meter per HDD). This is 39% of the national average of 15 Btu per square foot per HDD and in the top 15% of all single-family homes in North America (see “Comparing Your Home’s Energy Use to Others,” www.homeenergy.org/consumerinfo/benchmarking-energy-usage.php).

Electrical usage does contribute to home heating. Considering both electricity and natural gas, the 1959 test home has an energy usage of 9.1 Btu per square foot per HDD. In 2005, Michigan’s Habitat for Humanity built several Energy Star homes. The Habitat homes averaged a total energy (natural gas and electricity) usage of 8.5 Btu per square foot per HDD (see “Energy Star Homes in Michigan,” *HE* May/June ’07, p. 40).

Observations

By design, an ODWH can produce the design flow of hot water indefinitely. The Paloma Waiwela is rated to supply 7.4 gallons (28 liters) per minute with a temperature increase of 45°F (25°C). Units of this size are advertised as producing sufficient hot water to supply three appliances simultaneously. However the fan coil counts as approximately two appliances. Thus there may occasionally be times when demand exceeds supply. This has



The Paloma Waiwela PH28CIFS is a typical on-demand water heater.

occurred twice since fall 2005, but the delay lasted only minutes (the time it took waiting for the dishwasher to fill before the resident could take a shower) rather than hours (the time it would have taken waiting for a hot water tank to heat more water).

In circumstances where a single 200,000 Btu per hour ODWH provides insufficient space heating and hot water (as may happen in very large homes or commercial buildings), additional ODWHs can be added in parallel or separately. This expanded unit will still be less expensive than a conventional system with multiple furnaces or larger boilers.

Appliances that require water hotter than 125°F (52°C), such as dishwashers and sanitizing washing machines, will take longer to complete their cycle, since they must heat the water to the required temperature.

Numerous contractors made negative and condescending comments when they were told about this system, or when they observed it during installation. I attribute their comments to ignorance of technologies that they don’t normally work with, and to fear of the unknown. To address the most common comment, “This violates the building code,” my response was always, “What section of the building code does it violate?” In Alberta, Canada, the

Table 2. ODWH Seasonal Adjustments

Average Outdoor Temperature	Approximate ODWH Setting
above 50°F	125°F (50°C)
14°F to 50°F	140°F (60°C)
-22°F to 14°F	150°F (65°C)
below -22°F	160°F (70°C)

A home’s ODWH setting may vary because of home size and insulation rating.

answer is none! Yes, this configuration is different, but different does not mean wrong. Since building codes vary from region to region, it is important to verify compliance in your area.

A dual water- and space-heating system is capable of providing hot water for DHW and space heating to a residential home in Edmonton, Alberta, Canada, year-round, including winter periods with temperatures of -40°F (-40°C). This novel system replaced both the conventional furnace and the conventional hot water tank with a single high-efficiency appliance. With a total cost of \$3,600, this system is about half the price of separate high-efficiency appliances.

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For more information:

For the Okaloosa Gas District study comparing hot water tanks and ODWHs, go to www.okaloosagas.com/appliances/waterHeaters/waterheatertest.cfm.